



مستشفى ابن سينا للجراحات التخصصية
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WHOLE BRAIN VOLUMETRY IN ADULTS WITH EARLY ONSET, MRI-NEGATIVE EPILEPSY

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I have no disclosures



Introduction

- ❑ Individuals with epilepsy may have normal MRI scans, a condition known as MRI-negative epilepsy, representing a challenge to epilepsy surgery.
- ❑ Segmental volumetric changes can be used to identify specific brain regions that are affected by seizures and may correlate with epileptogenic focus lateralization and localization offered by EEG and seizure semiology.



Purpose

We aim at using whole brain morphometry to describe global and segmental volumetric changes at various cortical, subcortical, and white matter structures in a cohort of adult patients with early-onset, MRI-negative epilepsy.



Patients & Methods

Twenty patients (11 females, 9 males) were retrospectively included (mean age: 34.1 years $\pm$ 13.7). They have recent onset (<6 months) focal epilepsy and normal MRI findings on 3T.

Volumetric analysis (voxel-based morphometry) using the tool: "AI-Rad Companion Brain MR for Morphometry, 2019" by *Siemens Healthineers*. The FDA-approved tool shows volume changes relative to a normative database.

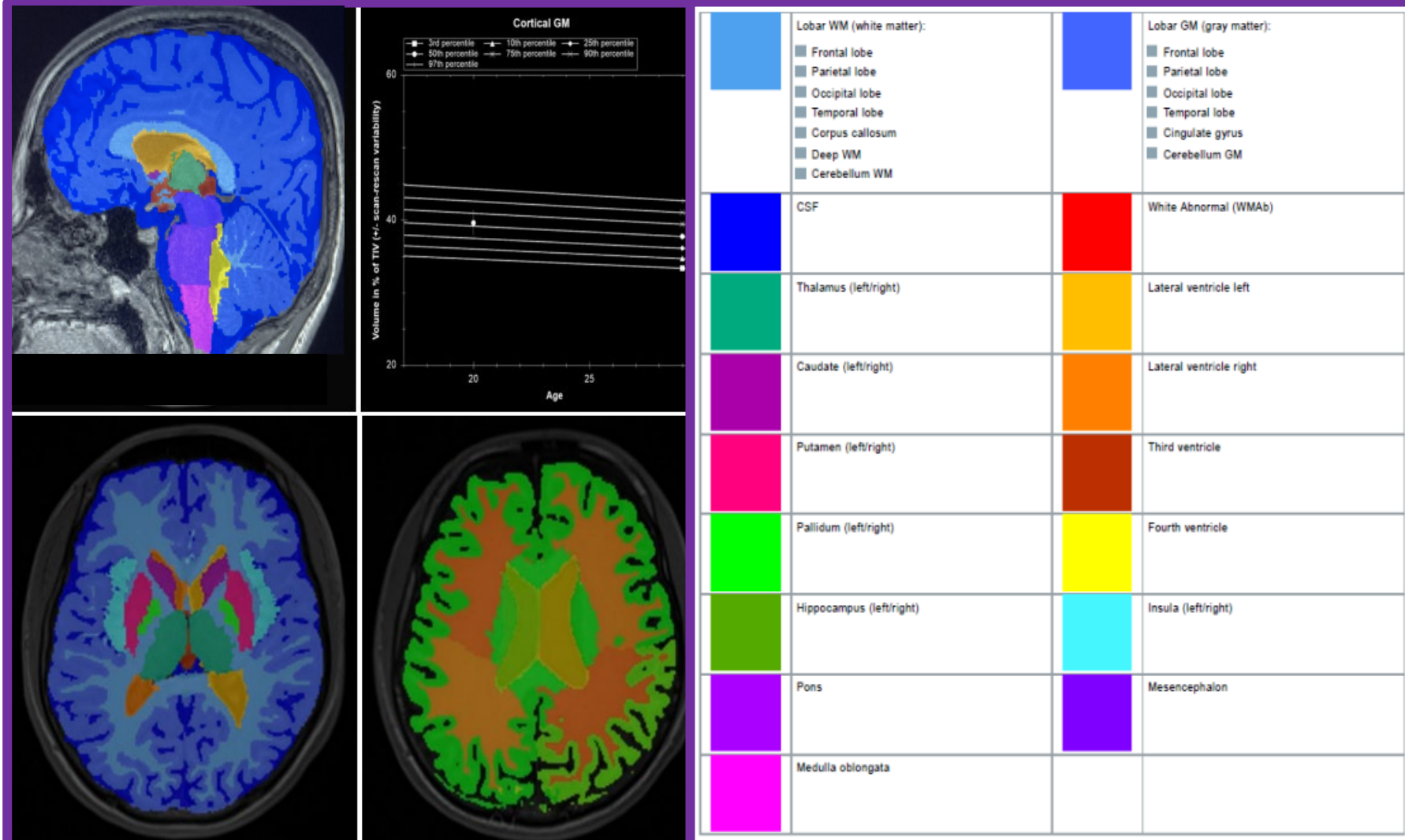
Review of semiology and a second reading of ictal/interictal EEG and MRI were independently performed by experienced epileptologist and neuroradiologist.



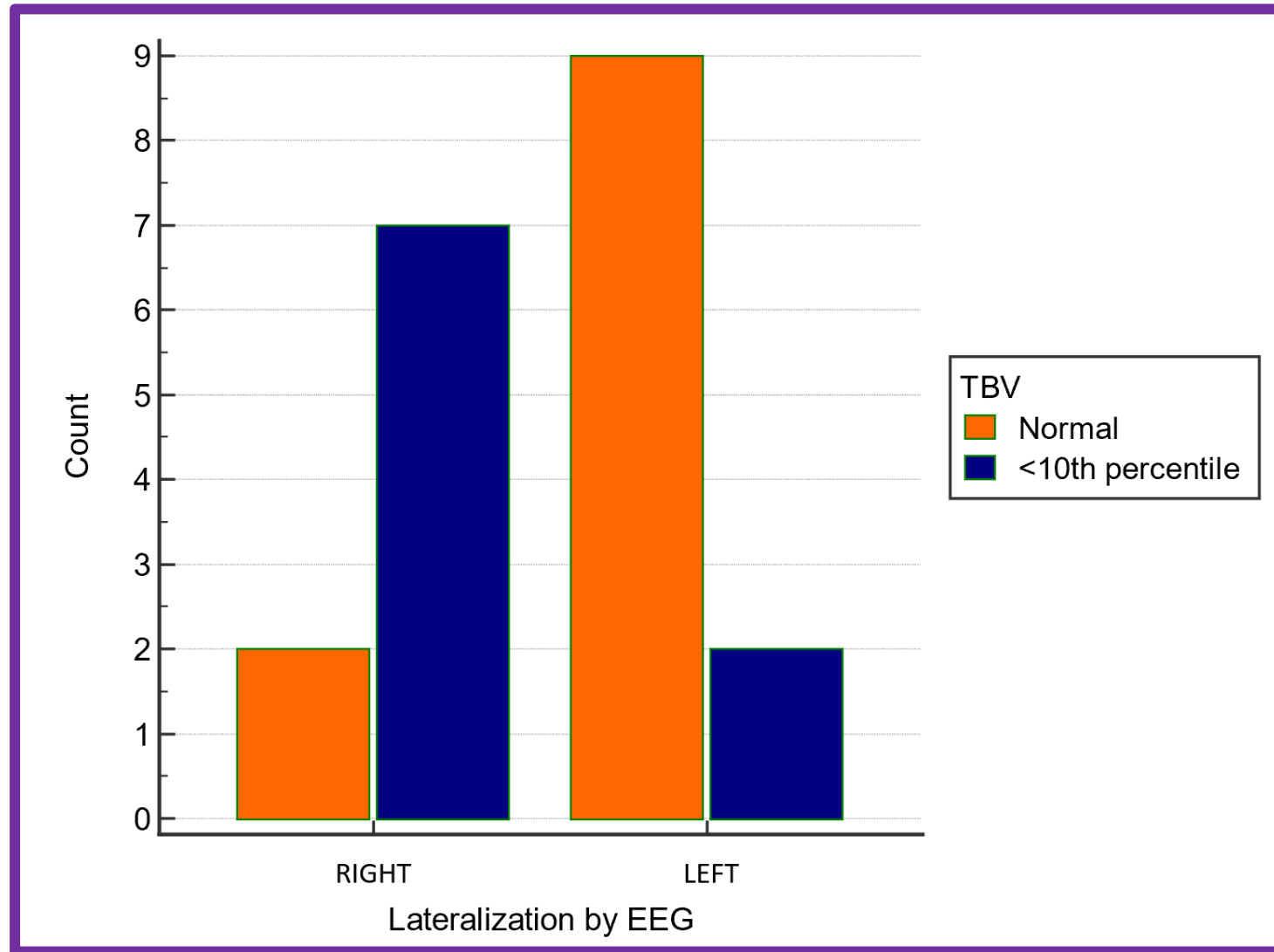
Volumetry tool: AI-Rad Companion Brain MR for Morphometry

Left: Anatomical T1 MPRAGE at the level of basal ganglia with corresponding segmentation color maps of GM and WM structures (bottom). Each volume is plotted against the percentiles of age/gender matched controls (top).

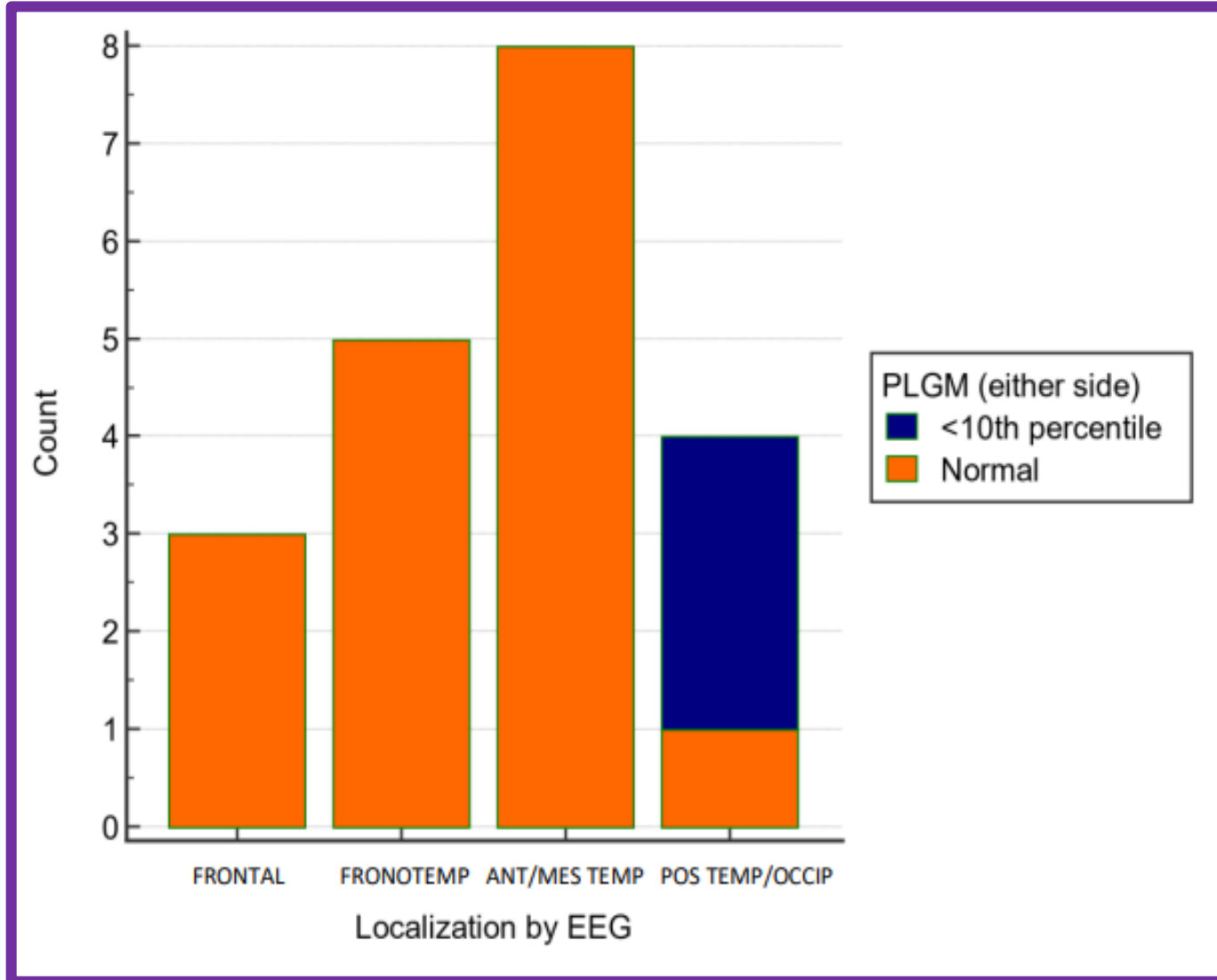
Right: Example of a color labelled key of segmented brain structures.



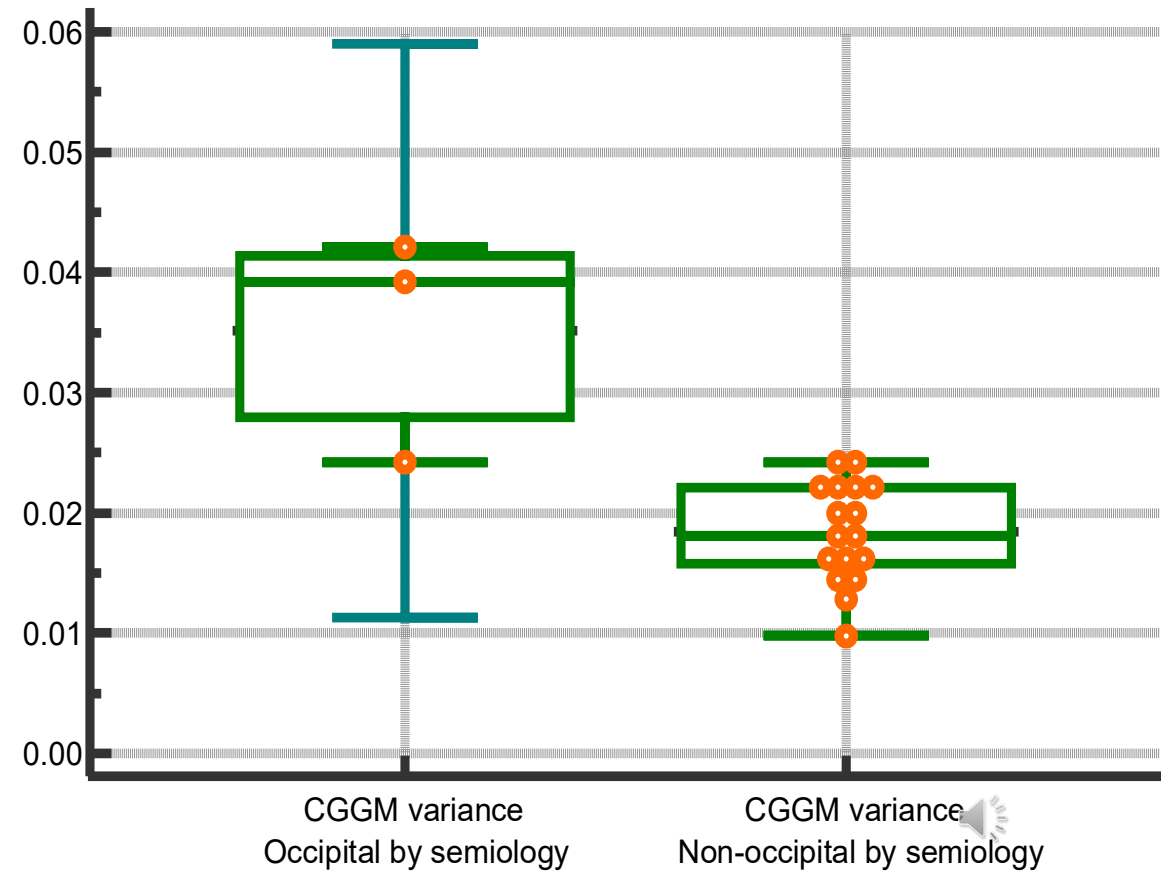
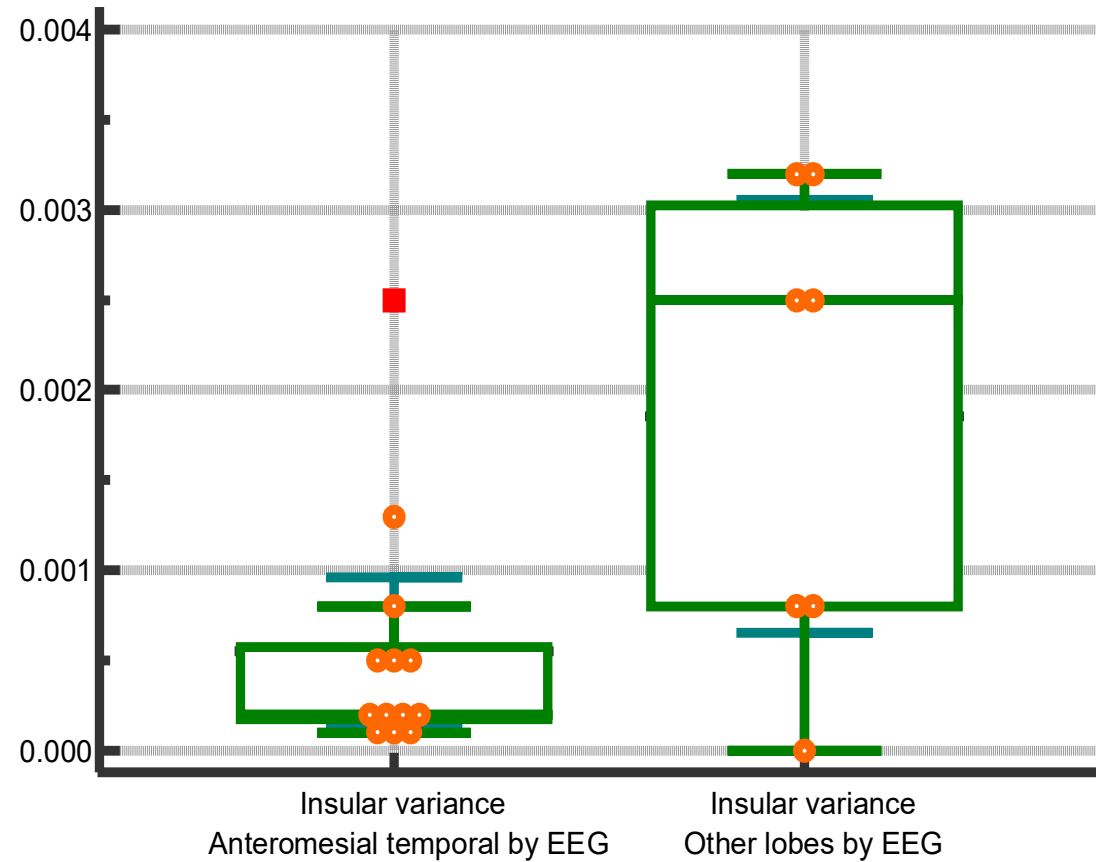
RESULTS (1): Total brain volume (TBV) reduction was more extensive (<10th percentile) in patients with right-sided (77.8%) versus left-sided (18.2%) epileptogenic activity, $p = 0.02$



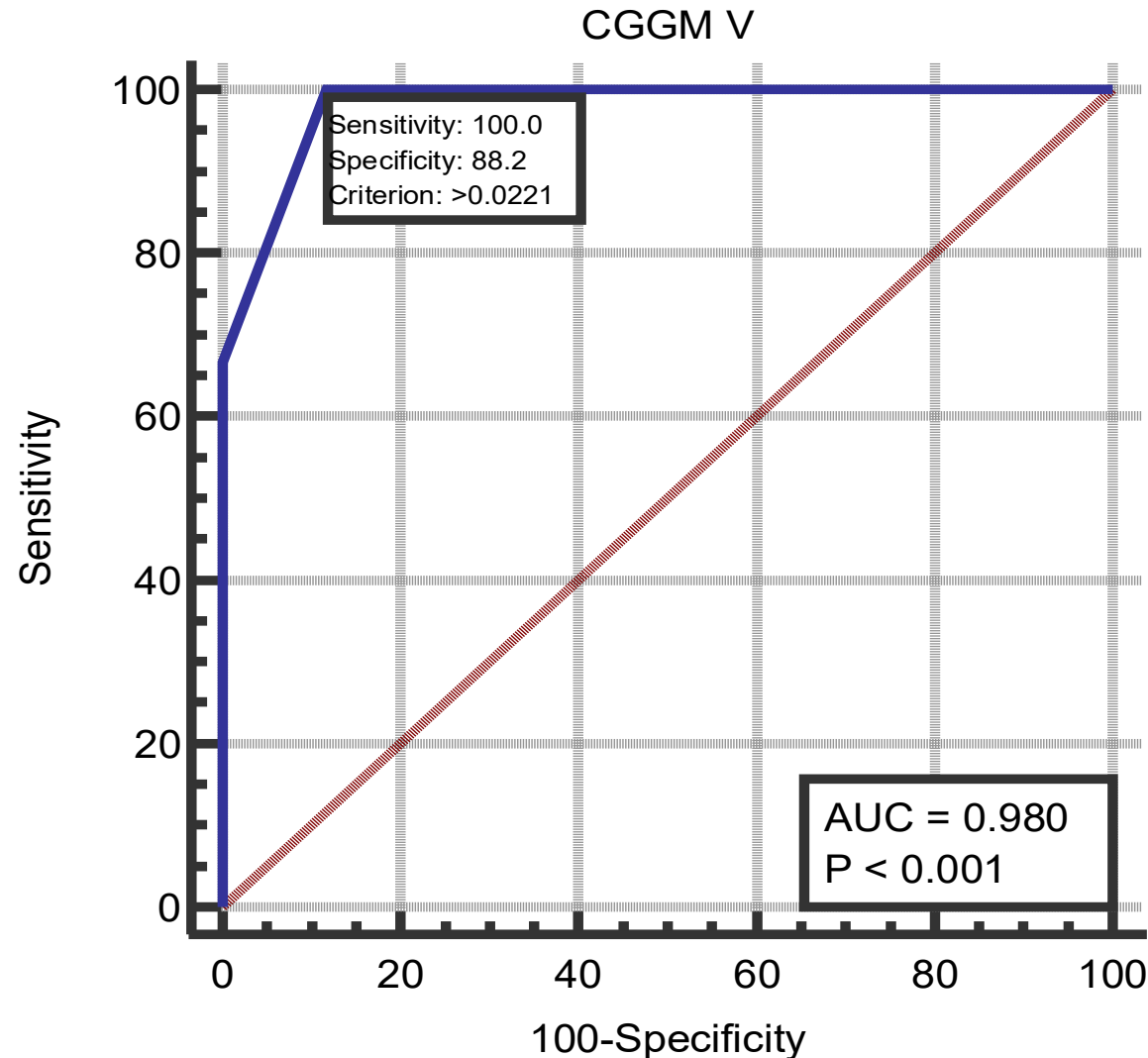
RESULTS (2): Reduced parietal lobe gray matter (PLGM) volume in 75% of patients with EEG-localized posterior temporal/occipital discharges while it was average in other groups, $p = 0.003$ (Bonferroni corrected)



RESULTS (3): Right-Left asymmetries of the insula ($p= 0.03$) and cingulate gyrus gray matter (CGGM) ($p= 0.009$) were associated with anteromesial temporal epilepsy by EEG and occipital epilepsy by semiology, respectively

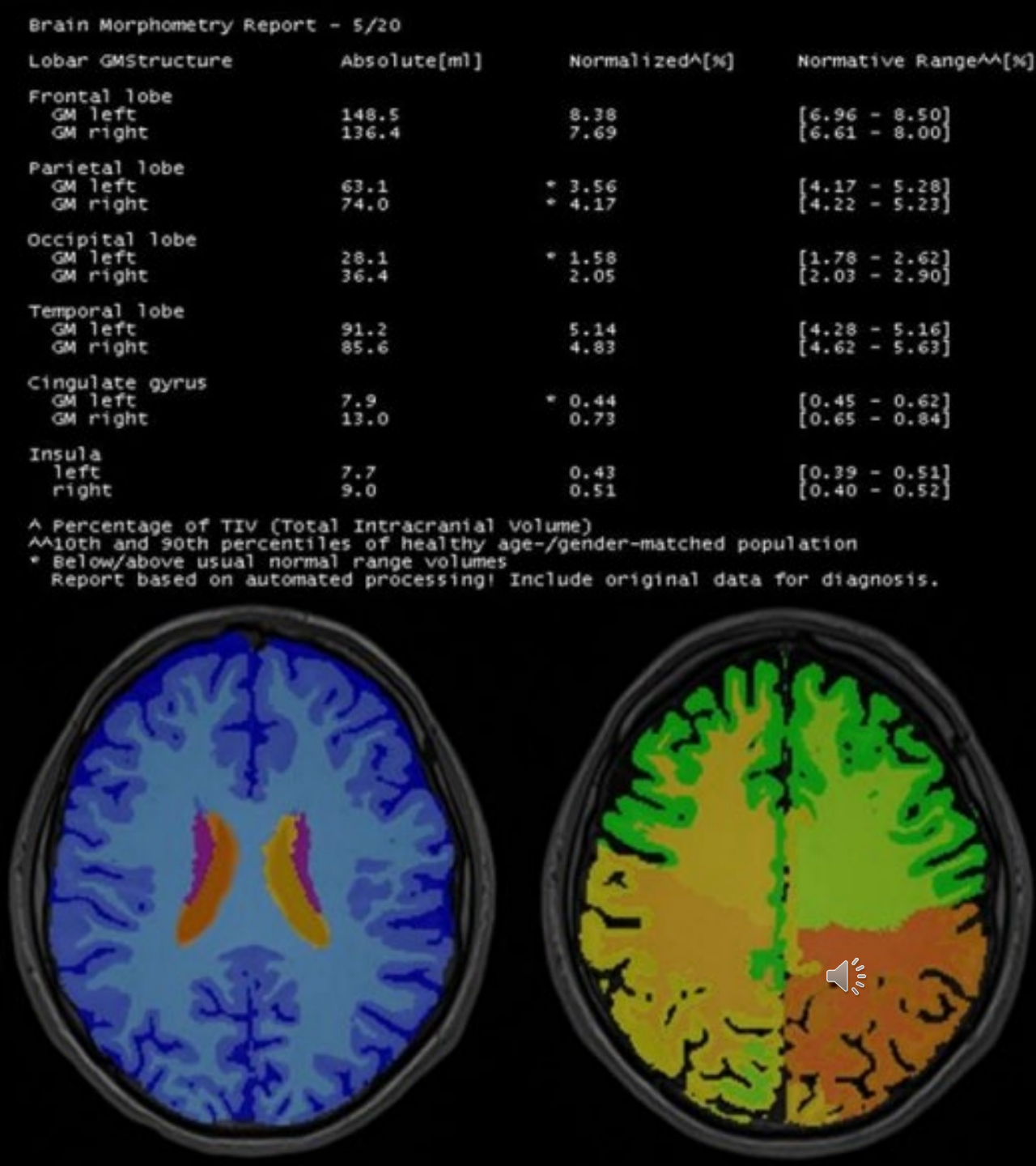


RESULTS (4): A right-left variance of (>0.022) of cingulate gyrus gray matter (CGGM V) normalized to intracranial volume is 88% sensitive, 100% specific in predicting semiology-defined occipital lobe epilepsy. AUC= 0.98, $p < 0.001$



Example case:

- 18 y/o patient diagnosed with occipital lobe epilepsy and presented with visual aura, poor color recognition, -ve MRI.
- Volumetry: reduced GM volumes (asterisk) below the 10th percentile of bilateral parietal lobes, left occipital lobe, and left cingulate gyrus, as well as cingulate gyrus (R-L) asymmetry.
- EEG: left occipital lobe discharges.



Discussion



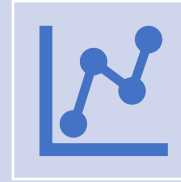
Patterns of volume loss in the current study are generally supported by the literature (ENIGMA study).



Volumetric changes are explained due to propagation of discharges from one cortical area to another.



Epilepsy is a disease that affects both white and gray matter. White matter loss (i.e., temporal lobes, pons) has been also reported in our study.



Technical limitations: retrospective design, the small sample size, the fully automated segmentation, using normative dataset versus the need of a control group. 🔊

Conclusion:

Our findings should promote further research on a larger number of epilepsy patients to explore the clinical value of volumetric changes including right-left asymmetries in potentially revealing subtle pathology in MRI-negative scans.



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Thank you

